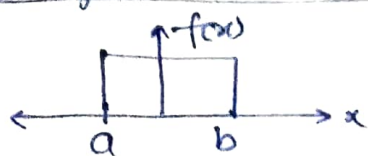


Uniform Distribution (Continuous Random Variable)



$$f(x) = \begin{cases} \frac{1}{b-a} & ; a < x < b \\ 0 & ; \text{otherwise} \end{cases} \quad \left. \vphantom{\begin{matrix} \frac{1}{b-a} \\ 0 \end{matrix}} \right\} \text{density function}$$

$$E(x) = \text{Mean}(\mu) = \frac{a+b}{2}, \quad \text{Variance} = \sigma^2 = \frac{(b-a)^2}{12}$$

$$\text{and Mean square value (MSV)} = \frac{a^2 + b^2 + ab}{3} \quad E(x^2)$$

$$E(X^n) = \int_{-a}^{\infty} x^n f(x) dx, \quad \text{Ex: } E(X^3) = \int_0^1 x^3 (1) dx$$

value of $f(x)$ and range given in question. $= \frac{1}{4}$

$$E(X^3) = \int_a^b x^3 \left(\frac{1}{b-a} \right) dx$$

Exponential Distribution (Continuous Random Variable)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

$$\text{Mean} = \frac{1}{\lambda}$$

$$\text{Variance} = \frac{1}{\lambda^2}$$

Since $x \geq 0$ always, so it's unilateral & it has only one parameter as ' λ '.

Gaussian Distribution (Discrete Random Variable)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} ; x \geq 0$$

There are two parameters σ, μ . $\left| \begin{array}{l} \mu \rightarrow \text{mean} \\ \sigma \rightarrow \text{SD} \end{array} \right.$

→ for continuous random variable we use integration and for discrete random variable we use Σ (summation).

Poisson Distribution (Discrete Random Variable)

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x = 0, 1, 2, \dots$$

$\lambda = \text{mean} = n \cdot p$ where; $n = \text{No. of trials}$, $p = \text{prob. success}$
 $x = \text{no. of successful trials}$

→ Base of x and λ should be same
if n is large → Poisson Distribution
small → Binomial

→ Mean = Variance = μ , $SD = \sqrt{\mu}$

Mean can be said as expectation, average, weighted average, first moment about origin.

Binomial Distribution: [Discrete Random Variable]

$$P(x) = {}^nC_x p^x q^{n-x}$$

n → no. of trials, p = probability of success,

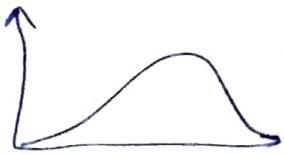
q = prob. of failure, x = no. of favourable counts.

Mean (μ) = np , Variance (σ^2) = npq

* Mode = 3median - 2mean

* ① for +ve skewed  Mode $\leq$ Median $\leq$ Mean

② Symmetric  Mode = Median = Mean

③ (-ve) skewed  Mean $\leq$ Median $\leq$ mode

* Coefficient of variation = $\frac{\sigma}{\mu}$

for continuous variable its probability density function given by,

① $f(x) \geq 0$ ② $\int_{-\infty}^{\infty} f(x) dx = 1$ ③ $\int_a^b f(x) dx$

for discrete random variable it is probability mass function.

1. $p(x) = P[X=x]$ 2. $p(x) \geq 0$ 3. $\sum p(x) = 1$

→ (Change of origin): Any constant added or subtracted than the SD remains unchanged but mean gets changed.

→ with change of scale all the 3 SD, mean and variance get changed.

→
$$I = \frac{1}{\sqrt{2\pi\sigma^2}} \int_0^\infty e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

for continuous random variable $\mu = E(x)$ then,
Standard Deviation = $E(x^2) - [E(x)]^2$

→
$$Z_{score} = \frac{\text{Normalized} - \text{Mean}}{SD} = \frac{x - \mu}{\sigma}$$

→ Normal distribution Curve 

→ Collectively exclusive $P(A \cup B) = P(S) = 1$

→ for standard Normal distribution $\mu = 0, \sigma^2 = 1$.

→ Second Moment = $\lambda^2 + \lambda$
first moment = mean = λ $\leftarrow$ Poisson Distribution.

Integration and Differentiation formulae

$$f'(x^n) = nx^{n-1}$$

$$f'(x \pm y) = \frac{d}{dx}(x) \pm \frac{d}{dx}(y)$$

$$f'(uv) = u f'(v) + v f'(u)$$

$$f'(u/v) = \frac{v f'(u) - u f'(v)}{v^2}$$

$$\frac{d}{dx} e^x = e^x$$

$$f'(a^x) = a^x \log a$$

$$f'(\log x) = 1/x \times (\log_e e)^{-1}$$

$$f'(\log_a x) = \frac{1}{x} \log_a e$$

$$f'(\sin x) = \cos x, f'(\cos x) = -\sin x$$

$$f'(\tan x) = \sec^2 x, f'(\cot x) = -\csc^2 x$$

$$f'(\sec x) = \sec x \tan x$$

$$f'(\csc x) = -\csc x \cot x$$

$$f'(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, f'(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$f'(\tan^{-1} x) = \frac{1}{1+x^2}, f'(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$f'(\sec^{-1} x) = \frac{1}{x \sqrt{x^2-1}}, f'(\csc^{-1} x) = \frac{-1}{x \sqrt{x^2-1}}$$

$$I(x^n) = \frac{x^{n+1}}{n+1} + C$$

$$I(\cos x) = \sin x + C$$

$$I(\sin x) = -\cos x + C$$

$$I(\sec x) = \tan x + C$$

$$I(\csc x) = -\cot x + C$$

$$I(\sec x \tan x) = \sec x + C$$

$$I\left(\frac{1}{1-x^2}\right) = \sin^{-1} x$$

$$I(e^x) = e^x + C, I(1/x) = \log x$$

$$I(a^x) = \frac{a^x}{\log a} + C$$

$$I(\tan x) = \log |\sec x| + C$$

$$I(\cot x) = \log |\sin x| + C$$

$$I(\sec x) = \log |\sec x + \tan x| + C$$

$$I(\csc x) = \log |\csc x - \cot x| + C$$

$$I\left(\frac{1}{x^2-a^2}\right) = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$I\left(\frac{1}{a^2-x^2}\right) = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$$

$$I\left(\frac{1}{x^2+a^2}\right) = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$I\left(\frac{1}{\sqrt{x^2-a^2}}\right) = \log |x + \sqrt{x^2-a^2}| + C$$

$$I\left(\frac{1}{\sqrt{a^2-x^2}}\right) = \sin^{-1} \frac{x}{a} + C$$

$$I\left(\frac{1}{\sqrt{x^2+a^2}}\right) = \log |x + \sqrt{x^2+a^2}| + C$$

Limits formulas

$$\lim_{x \rightarrow 0} \sin x = 0$$

$$\lim_{x \rightarrow 0} \cos x = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \text{ [Max Value]}$$

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

$$\lim_{x \rightarrow 0} e^x = 1$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{a}{x}\right)^x = e^a$$

$$\text{If } \lim_{x \rightarrow \infty} f(x) = \infty$$

then

$$\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$$

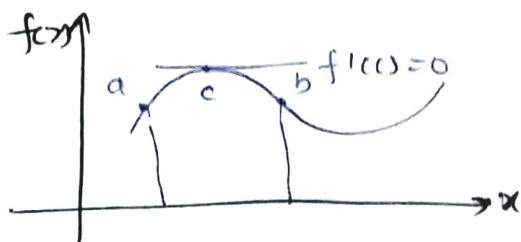
$$m_1 m_2 = -1$$

Mean Value Theorem

Rolle's Theorem : Let $f(x)$ is defined in $[a, b]$ such that it satisfies 3 conditions.

- i) $f(x)$ is continuous in $[a, b]$ [No holes]
- ii) $f(x)$ is differentiable in (a, b) [No sharp point]
- iii) $f(a) = f(b)$.

then, $\exists$ atleast one point $c \in (a, b)$ such that $f'(c) = 0$



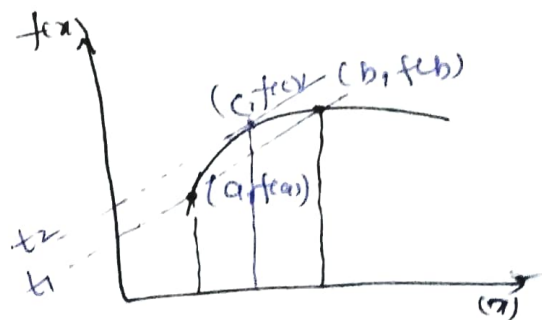
Lagrange's mean value : $f(x)$ is defined in $[a, b]$ such that it satisfies 2 conditions.

- i) $f(x)$ is continuous in $[a, b]$
- ii) $f(x)$ is differentiable in (a, b) then there exist atleast one point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\text{slope of } t_1 = \frac{f(b) - f(a)}{b - a}$$

if t_1 and t_2 are parallel then $f'(c) = \frac{f(b) - f(a)}{b - a}$



Cauchy's Mean Value Theorem

Let $f(x)$ and $g(x)$ be defined in $[a, b]$ such that

- i) $f(x)$ and $g(x)$ are continuous in $[a, b]$
- ii) $f(x)$ and $g(x)$ are differentiable in (a, b)
- iii) $g'(x) \neq 0 \forall x \in (a, b)$ then $\exists$ atleast one point $c \in (a, b)$, such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$$

$$\left| \begin{aligned} f'(c) &= \frac{f(b)-f(a)}{b-a} \\ g'(c) &= \frac{g(b)-g(a)}{b-a} \end{aligned} \right.$$

Q The mean value 'c' of $f(x) = e^x [\sin x - \cos x]$ in the interval $[\frac{\pi}{4}, \frac{5\pi}{4}]$ is.

Since, $f(\frac{\pi}{4}) = f(\frac{5\pi}{4}) = 0$ so we can use Rolle's theorem,

$$f'(x) = 2e^x \sin x$$

So, for $c \in (\frac{\pi}{4}, \frac{5\pi}{4})$ so that $f'(c) = 0$

$$\Rightarrow 2e^c \sin c = 0$$

$$\Rightarrow \sin c = 0 \quad \text{So, } c = 0, \pm\pi, \pm 2\pi, \dots$$

So, $c \in (\frac{\pi}{4}, \frac{5\pi}{4})$ and also $c \nearrow$

$$\text{So } \boxed{c = \pi} \text{ only}$$

② $f(x) = x(x+3)e^{-x/2}$, find mean value 'c' in interval $[-3, 0]$

$f(-3) = f(0) = 0$ so we apply Rolle's theorem

After procedure,

→ Differentiable function is always continuous.

$$c = 3, -2 \text{ but only } -2 \in [-3, 0]$$

$$\text{So, } \underline{\underline{c = -2}}$$

③ The mean value 'c' for $f(x) = \sqrt{x^2 - 4}$ in $[2, 4]$

$$f(a) \neq f(b)$$

So we use mean value theorem

$$f'(c) = \frac{c}{\sqrt{c^2 - 4}} = \frac{f(b)-f(a)}{b-a} \Rightarrow \frac{c}{\sqrt{c^2 - 4}} = \frac{\sqrt{12}}{2} = \sqrt{3}$$

$$\Rightarrow 2c^2 = 12 \Rightarrow \underline{\underline{c = \pm\sqrt{6}}} \text{ but only } +\sqrt{6} \in [2, 4].$$

Eigen Vectors : find eigen values first then,
 using $|A - \lambda I| = 0$

⇒ Sum of eigen values = trace

⇒ determinant of matrix from characteristic eqn.

If eqn is $a\lambda^3 + b\lambda^2 + c\lambda + d = 0$ then

$|A| = d$ (constant term)

Ex: $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ $[A - \lambda I][X] = 0$

Sol: Eigen values are $\lambda_1 = 0, \lambda_2 = 3, \lambda_3 = 15$ then
 eigen vectors are:-

$[A - \lambda I]$ at $\lambda = 0$ is $\begin{bmatrix} 8 & -6 & 2 \\ 6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 8 & -6 & 2 \\ 6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $\begin{bmatrix} 8 & -6 & 2 \\ 6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ → Determinant

⇒ $\frac{x_1}{5} = \frac{-x_2}{-10} = \frac{x_3}{10} = \frac{x_4}{1} = \frac{x_2}{2} = \frac{x_3}{2} = x \begin{bmatrix} 1 \\ 2 \\ 2 \\ 2 \end{bmatrix}$

→ Determinant of 4×4 Matrix :

$\begin{vmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{vmatrix} = 2 \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix}$
 $- 1 \begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = 5$] Alternate signs
 (+), (-), (+), (-).

2nd Approach: Trick

→ No trick

Day-4

Calculus.

(51)

→ Read all the differentiation and Integration formula from the main notes.

$$\textcircled{1} \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{x^2}{2}} = \frac{1}{2}$$

→ In $\sin(x)$, put the value in radian or angle accordingly.

→ If there exist a 'y' in interval (a,b) then according immediate value theorem verify the options.

→ Various limits forms:-

① $\frac{0}{0}$ or $\frac{\infty}{\infty}$: Apply L-Hospital Rule.

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{(1 - \cos ax)}{x^2} = \frac{a^2}{2}$$

③ for $\frac{\infty}{\infty}$ form take x as common like $\frac{x(1 + 5/x)}{x}$.

④ 0^0 form take log on both sides.

~~⑤~~ 1^∞ : $\lim_{x \rightarrow 0} f(x)^{g(x)} = e^{\lim_{x \rightarrow 0} g(x)[f(x)-1]}$

$$\textcircled{6} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) / \left(\frac{\tan x}{x} \right) / \left(\frac{x}{\sin x} \right) = 1$$

→ In Rolle's theorem $f(a) = f(b)$

→ In Lagrange's theorem $f(a) \neq f(b)$.

→ In Cauchy mean value theorem $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$.

→ Theorem: The max. value of $(x^T A x)$ where the max. is taken over all 'x' that are eigen vector of A is the max. eigen value of A.

→ Eigen values of upper triangular is diagonal elements.

~~→~~ If eigen value of A is $x_1, y_1, \dots$ then for A^n it becomes $x_1^n, y_1^n, \dots$.

The value of dot product of ~~any~~ the eigen vectors corresponding to any pair of different eigen values of any symmetric positive definite matrix is 0.

→ Idempotent: $A^2 = A$ | → $(AB)^T = B^T A^T$

→ Involuntary: $A^2 = I$ | → $(AB)^0 = B^0 A^0$

→ Symmetric: $A = A^T$, $A + A^T/2$ | : Orthogonal → $A A^T = I$

→ Skew Sym: $A^T = -A$, $A - A^T/2$ | → $A^0 = A^T$: Unitary

→ $A^{-1} = \frac{1}{|A|} (\text{Adj } A)$, $|AB| = |A| |B|$, $|A^n| = |A|^n$

$|\text{adj}(A)| = |A|^{n-1}$, $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$, $|A^T| = |A|$

→ $(AB)^T = B^T A^T$; $(A^T)^T = (A^T)^T$, $A A^T = I$

→ $\text{rank}(AB) \leq \text{rank}(A)$ or $\text{rank}(AB) \leq \text{rank}(B)$

$\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$

If $\text{rank}(A) = \text{no. of vectors}$: linearly independent

$\text{rank}(A) < \text{no. of vectors}$: ——— dependent

→ To find eigen vector use $(A - \lambda I)X = 0$ and to verify use $AX = \lambda X$.

→ LU decomposition is important

→ Complex eigen values occurs in pairs.

→ Determinant = product of eigen values

→ If $|A| = 0$ it means $\text{rank}(A) < n$ (or less)

→ A matrix can be diagonalized iff it has all the different eigen values or all the rows are independent, $\text{rank} = n$.

→ Remember Cayley Hamilton theorem

Probability

(S2)

→ Conditional Probability: $P(F/E) = \frac{P(F \cap E)}{P(E)}$
It means probability of (F) given that (E) have been already occurred.

→ Multiplication Theorem:

$$P(A \cap B) = \frac{P(B) P(A/B)}{P(A)} = \frac{P(A) P(B/A)}{P(B)}$$

$$\text{or } P(E_1 \cap E_2 \cap E_3) = P(E_1) * P(E_2/E_1) * P(E_3/E_1 \cap E_2)$$

→ Independent Events: $P(A/B) = P(A)$, $P(B/A) = P(B)$
or $P(A \cap B) = P(A) * P(B)$

→ Baye's Theorem: $P(A/R) = \frac{P(A) P(R/A)}{P(A) P(R/A) + P(B) P(R/B) + P(C) P(R/C)}$

→ Bernoulli Distribution

$P(X=1) = p$ denotes success, $P(X=0) = (1-p)$ denote failure

$${}^nC_r (p)^r (1-p)^{n-r}$$

→ Random Variable: $S = \{HH, TH, HT, TT\}$

↳ $X = \{2, 1, 0\}$ → chance of getting head

$$P(X=2) : 1/4 ; P(X=1) = 1/2 ; P(X=0) : 1/4$$

→ Uniform distribution: (Continuous Random Variable)

$$f(x) = \begin{cases} \frac{1}{b-a} ; a < x < b \\ 0 ; \text{otherwise} \end{cases} \text{ density fn}$$

$$E(X) = \text{Mean}(\mu) = \frac{a+b}{2} , \text{ Variance } (\sigma^2) = \frac{(b-a)^2}{12}$$

$$E(X^2) : \text{mean square value} = \frac{a^2 + b^2 + ab}{3}$$

$$E(X^n) = \int_{-\infty}^{\infty} x^n f(x) dx \quad [\text{Range of Integration } a \rightarrow b]$$

Exponential Distribution: (Continuous Random Variable)

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases} \quad \begin{array}{l} \text{Mean} = \frac{1}{\lambda} \\ \text{Variance} = \frac{1}{\lambda^2} \end{array}$$

Since $x \geq 0$ (always). So its unilateral with one parameter λ .

Discrete Random Variable

① Gaussian Distribution: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} ; x \in \mathbb{R}$

$\mu \rightarrow$ mean, $\sigma \rightarrow$ standard deviation

$\rightarrow$ for continuous random variable we use integration and for discrete we use Σ (summation).

② Poisson Distribution: $f(x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x = 0, 1, 2, \dots$
 $\lambda = \text{mean} = n \cdot p$

Base of λ and x should be same
if $n \rightarrow$ large: poisson distribution
if $n \rightarrow$ small: binomial distribution

$x \rightarrow$ sample random variable

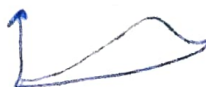
Mean = Variance = μ , SD = $\sqrt{\mu}$

③ Binomial Distribution: $P(x) = \frac{n!}{x! (n-x)!} p^x q^{n-x}$
Mean (μ) = np ; Variance (σ^2) = npq

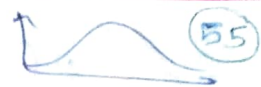
: Mode = 3 Median - 2 Mean

+ve skewed $\hookrightarrow$ Mode $\leq$ Median $\leq$ Mean

-ve skewed $\hookrightarrow$ Mean $\leq$ Median $\leq$ Mode



~~Symmetric~~ $\hookrightarrow$ Mode = Mean = Median



~~Coefficient of Variation~~ = σ/μ

for continuous variable its probability density fn is given by;

① $f(x) \geq 0$ ② $\int_{-\infty}^{\infty} f(x) dx = 1$ ③ $\int_a^b f(x) dx$

for discrete random variable it is probability mass fn: ① $P(x) = P[X=x]$ ② $P(x) \geq 0$ ③ $\sum P(x) = 1$

$\rightarrow$ With change of origin [add or subtract constant], SD remains unchanged but mean gets changed.

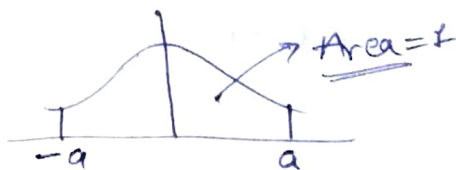
$\rightarrow$ With change of scale [SD, mean, variance] all gets changed.

$$\rightarrow I = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

~~9m~~ for continuous random variable if $\mu = E(x)$ then, Standard Deviation = $\sqrt{E(x^2) - [E(x)]^2}$

$$\rightarrow Z \text{ score} = \frac{\text{Normalized} - \text{Mean}}{\text{SD}} = \frac{x - \mu}{\sigma}$$

~~Normal Distribution Curve~~



$\rightarrow$ for standard Normal Distribution: $\mu = 0, \sigma^2 = 1$

$\rightarrow$ In Poisson Distribution: First Moment = mean = λ
Second Moment = $\lambda^2 + \lambda$

$\rightarrow$ If two fns are executing sequentially then total time taken = $\int_0^t f(x) f(t-x) dx$ for probability density fn.

$P(A \cup B) \leq P(A) + P(B)$

if 2 events are mutually exc. then $P(P \cap Q) = \phi$

$P(P \cap Q) \leq P(P) \text{ or } P(Q)$

There is no relation b/w mutually exclusive & independent events

→ Cumulative Distributed function is value upto 'x'

→ for normal distribution or gaussian random variable Mean = Median = Mode = 0

→ Solⁿ of Homogeneous System of Linear Equation.

: Inconsistent: (Not possible bcoz each matrix must have solⁿ as 0)

Consistent → Unique or trivial: $|A| \neq 0$, rank = order
 → Non-trivial: Infinite: $|A| = 0$, rank < order

Solⁿ for Non-Homogeneous:

$AX = B$ → Inconsistent; $r[A|B] \neq r[A]$

→ Consistent:
 → $r[A|B] = r[A] = n$ (Unique)
 → $r[A|B] = r[A] < n$ (Infinite)

→ Standard deviation (s) = $\sqrt{\frac{n \sum x_i^2 - (\sum x_i)^2}{n^2}}$

→ $E(kx) = k \cdot E(x)$, $\text{Var}[kx] = k^2 \text{Var}[x]$

$E(k-x) = k - E(x)$, $\text{Var}(k-x) = (-k)^2 \text{Var}[x]$

→ No. of distinct eigen values (max) = size of matrix

→ ~~No.~~ Sum of eigen value = trace of matrix

→ Eigen values of Hermitian matrix are real

- $\rightarrow (x-\alpha)(x-\beta) > 0$ then, $x \in (-\infty, \alpha] \cup [\beta, \infty)$
 $\rightarrow (x-\alpha)(x-\beta) < 0$ then, $x \in (\alpha, \beta)$ $\rightarrow$ open bracket
 $\rightarrow (x-\alpha)(x-\beta) = 0$ then, $x = \alpha, \beta$
 $\rightarrow$ if not continuous then it will not be differentiable but converse is not true.

In case of partial differential eqⁿ:

$$a = \frac{\partial f}{\partial x} = 0 \text{ to find } x$$

$$b = \frac{\partial f}{\partial y} = 0 \text{ to find } y$$

then,

$$r = \frac{\partial^2 f}{\partial x^2}, \quad t = \frac{\partial^2 f}{\partial y^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}$$

then, find $rt - s^2 = k$.

① if $k > 0$, then, (a, b) are points of maxima and $r < 0$

② if $k > 0$ and $r > 0$ then point of minima

$\rightarrow$ According to Baye's Theorem.

$$P(E_i/A) = \frac{P(E_i) P(A/E_i)}{\sum_{k=1} P(E_k) P(A/E_k)}$$

Basic Differentiation and Integration : Revise from main Notes

$\rightarrow$ Also revise all the trigonometric identities.

$$\rightarrow f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

→ If eigen values of matrix 'P' are $[a_1, a_2, \dots]$ then
 if $f(b) = p^3 + p^2 + 1$ or any function given then eigen
 values of $f(p)$ is $[f(a_1), f(a_2), \dots, f(a_n)]$

→ The measure of skewness is dependent upon
 amount of dispersions

$$\rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

→ Area of $\Delta ABC = \frac{1}{2} (\text{product of sides}) \times (\text{angle b/w them})$

→ for a given matrix A if eigen vector given is
 $\vec{V}$ and eigen value is λ then,

$$\boxed{A \cdot \vec{V} = \lambda \vec{V}}$$

→ Determinant of matrix = Product of eigen values

$$\rightarrow (A+B)^T = A^T + B^T, \quad (\lambda A)^T = \lambda (A^T)$$

$$\rightarrow \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ d_{21} & 1 & 0 \\ d_{31} & d_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

→ $(A - \lambda I)X = 0$ for getting eigen vector from
eigen values

$P(X \leq -1)$ and given that $\mu_x = -1, \sigma_x^2 (\text{variance}) = 1$
 then, $Z_x = \frac{-1 - \mu_x}{\sigma_x} = -\frac{2}{2} = -1$
 $P(Z_x \leq -1)$

$$\left[\frac{-1 - \mu_x}{\sigma_x} \right]$$